# Problem Set 4

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

**Relevant reading**: ILA 4.1–4.4, 6.1–6.2. Problems marked [**code**] should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from ILA, 5th edition (Strang, 2016).

1. ILA, Problem Set 4.2: problems 1, 2, 3, 4, 5, 6, 7, 10, 13, 17, 21, 22, 23, 28, 31, 32, *and* one extra credit problem: 34.
2. Let $\boldsymbol{a} \in \mathbb{R}^n$ be nonzero and let $\boldsymbol{b} \in \mathbb{R}^n$. Prove that the projection $\boldsymbol{p} = \frac{\boldsymbol{a}^\top \boldsymbol{b}}{\boldsymbol{a}^\top \boldsymbol{a}} \boldsymbol{a}$ is the unique point on the line through $\boldsymbol{a}$ closest to $\boldsymbol{b}$; that is, $\|\boldsymbol{b} - \boldsymbol{p}\| < \|\boldsymbol{b} - x\boldsymbol{a}\|$ for every scalar $x$ with $x\boldsymbol{a} \neq \boldsymbol{p}$.
3. Let $\boldsymbol{A}$ be an $m \times n$ matrix with orthonormal columns. Show that the matrix projecting onto the column space of $\boldsymbol{A}$ simplifies from $\boldsymbol{P} = \boldsymbol{A}(\boldsymbol{A}^\top \boldsymbol{A})^{-1} \boldsymbol{A}^\top$ to $\boldsymbol{P} = \boldsymbol{A}\boldsymbol{A}^\top$, so that no matrix inverse needs to be computed.
4. ILA, Problem Set 6.1: problems 2, 3, 4, 5, 6, 9, 10, 11, 19, 22, 30, 37.
5. [**code**] Let $\boldsymbol{a} = (2, 1)^\top$ and let $\boldsymbol{P}$ be the matrix that projects onto the line through $\boldsymbol{a}$. Build $\boldsymbol{P}$ in code and check numerically that $\boldsymbol{P}^\top = \boldsymbol{P}$ and $\boldsymbol{P}^2 = \boldsymbol{P}$. Compute the eigenvalues and eigenvectors of $\boldsymbol{P}$, and explain why the eigenvalues must be 1 and 0 and what their eigenvectors are geometrically. Finally, plot the line through $\boldsymbol{a}$ together with two or three vectors $\boldsymbol{b}$ of your choice and their projections $\boldsymbol{P}\boldsymbol{b}$, joining each $\boldsymbol{b}$ to $\boldsymbol{P}\boldsymbol{b}$ by a dashed segment, and confirm from the plot that $\boldsymbol{b} - \boldsymbol{P}\boldsymbol{b}$ is perpendicular to the line.
6. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).